

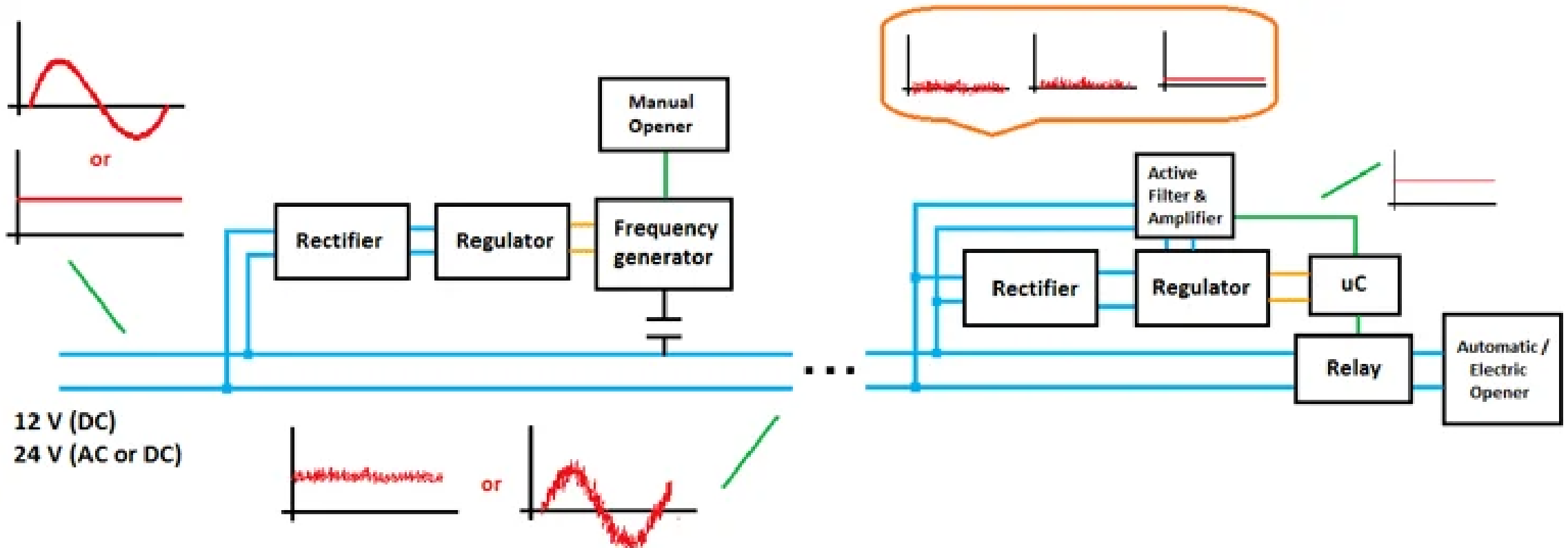
Power Line Communication

By

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Basic Principle:



M1 carrier telephone system



Challenges of PLC:

In the M1 carrier telephone system in 1925, Wolfe and co-authors identified three main development problems:

1) Understanding the line behaviour:

The high-frequency behaviour of power lines and related equipment had to be studied together with power companies.

2) Coupling to high-voltage lines:

A safe and efficient way was needed to connect carrier communication equipment to high-voltage conductors.

3) Developing the communication equipment:

Only after the line behaviour and coupling problems were understood could the actual communication circuits and devices be designed.

NETWORK STRUCTURE DATA

Circuit lengths 2025

Overhead power lines

Medium voltage	0.63	km
Low voltage	0.95	km

Cable lines

High voltage	8.84	km
Medium voltage	375.94	km
Low voltage	994.37	km

Total circuit lengths

Total length	1,380.73	km
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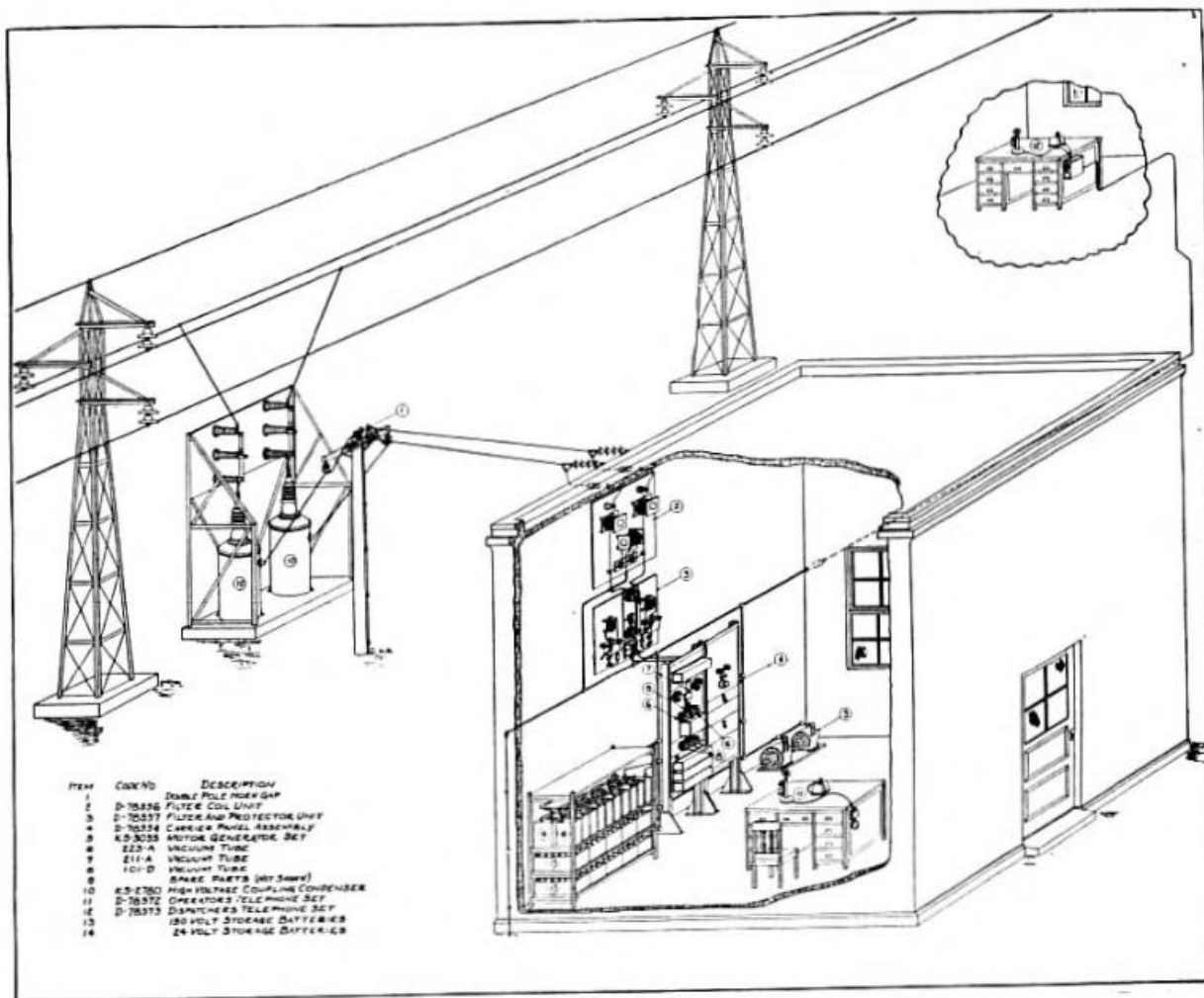


Fig. 15—Typical Layout of Power Line Carrier Telephone System, Using High Voltage Condensers for Coupling to Power Line

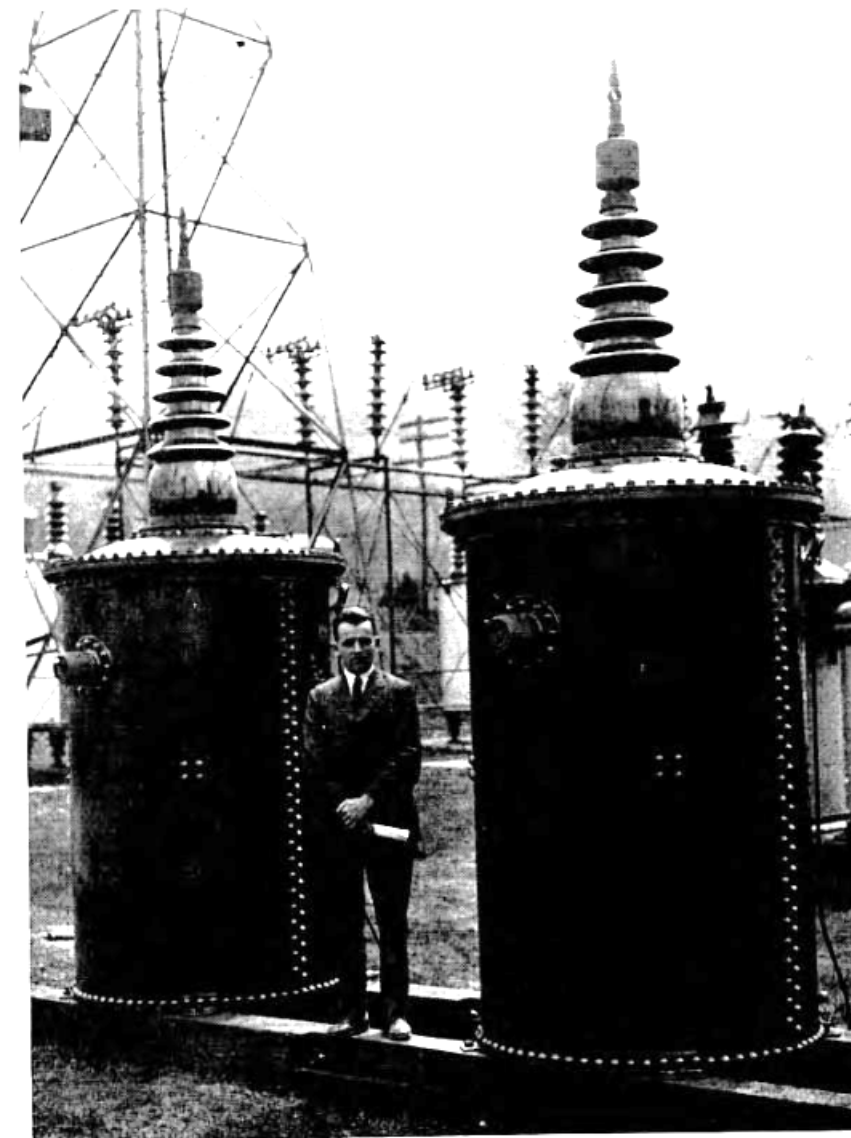


Fig. 13—110 K.V. Coupling Condensers Used for Coupling Carrier Circuit to a 110 K.V. Power Line

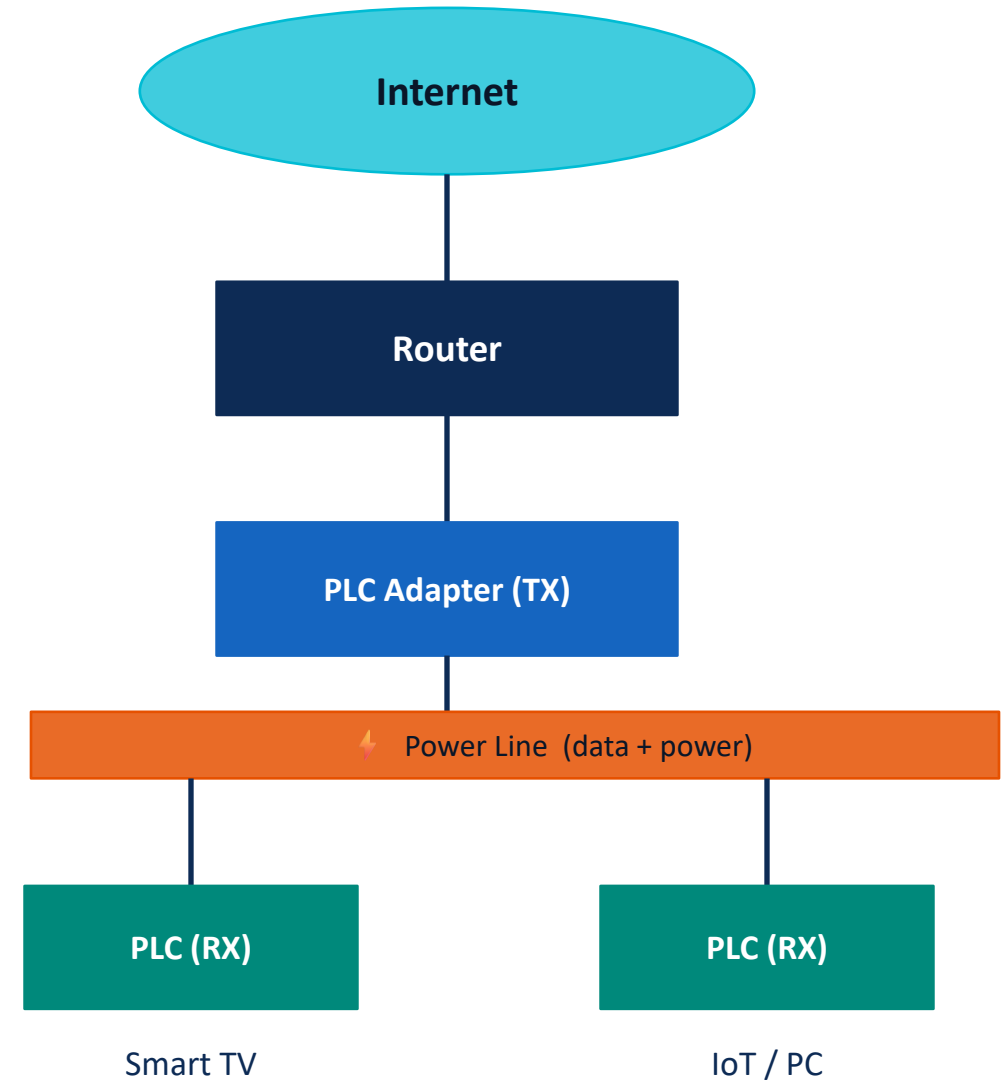
Home Applications



Residential PLC Network Architecture

How it works

- ① Router receives internet signal
- ② PLC adapter injects high-freq. data signal onto power wiring
- ③ Signal travels through household electrical cables
- ④ Filtering separates data from power at receiving end
- ⑤ Devices get network access — no new cables needed



PLC vs. Wi-Fi

PLC



<https://bit-electronix.eu/STRONG-POWERL1000WFDUOEUV2-Powerline-Adapter-1000-Mbit-s-HomePlug-AV2-mit-WLAN-Wi-Fi-5-Gigabit-Ethernet-Eingebaute-Steckdose-2er-Pack-Weiss/POWERL1000WFDUOEUV2-B?srsltid=AfmBOorKLFhGo9cOX4sqe6Q-D7m39huLvb1pVorQQd23vnpV79ju-zYB>

Wi-Fi



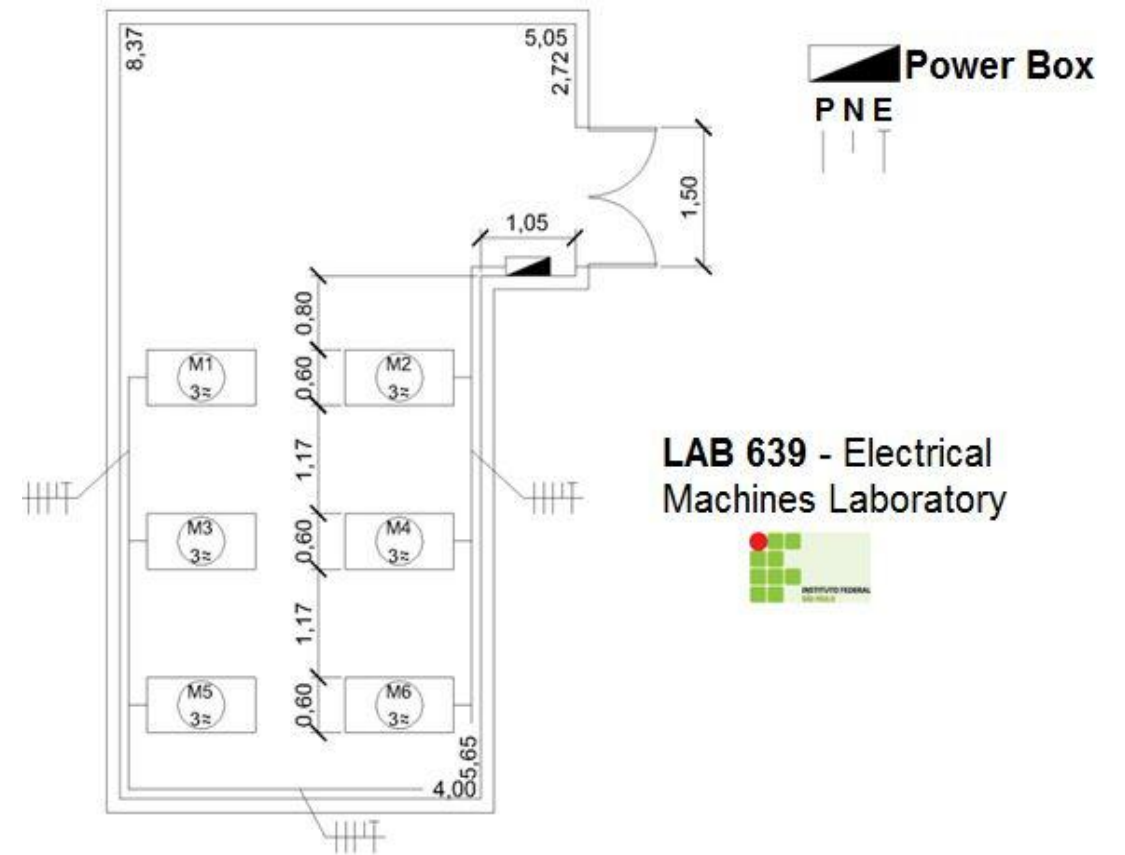
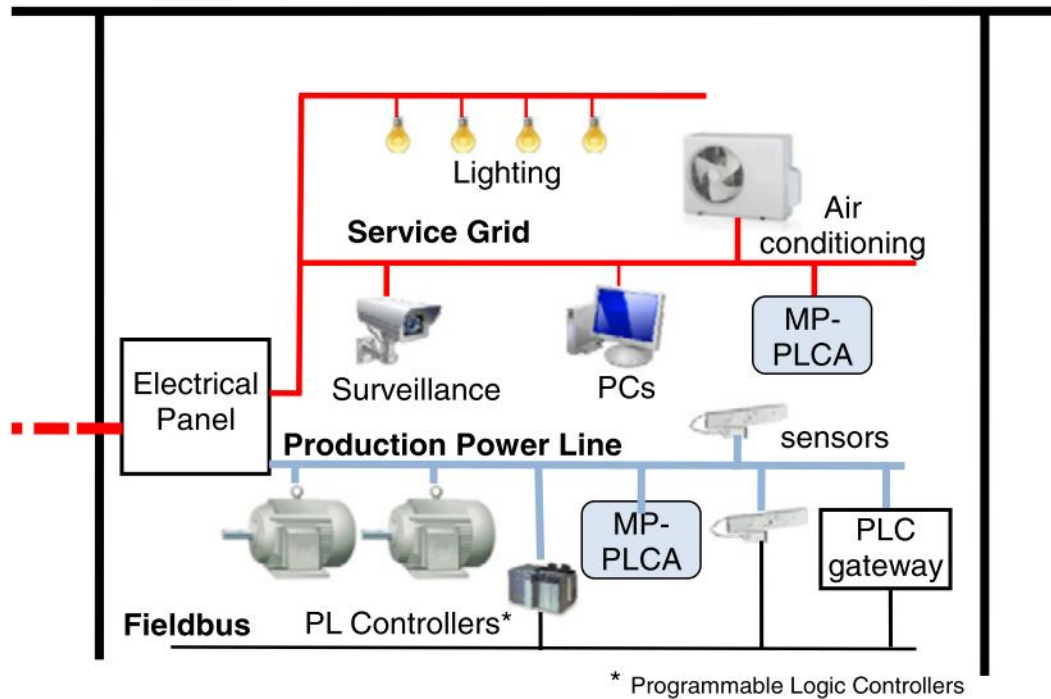
https://img.magnific.com/premium-photo/new-wifi-router-white-wooden-table-against-brick-wall-space-text_495423-56020.jpg?semt=ais_hybrid&w=740&q=80

Mixed-medium applications

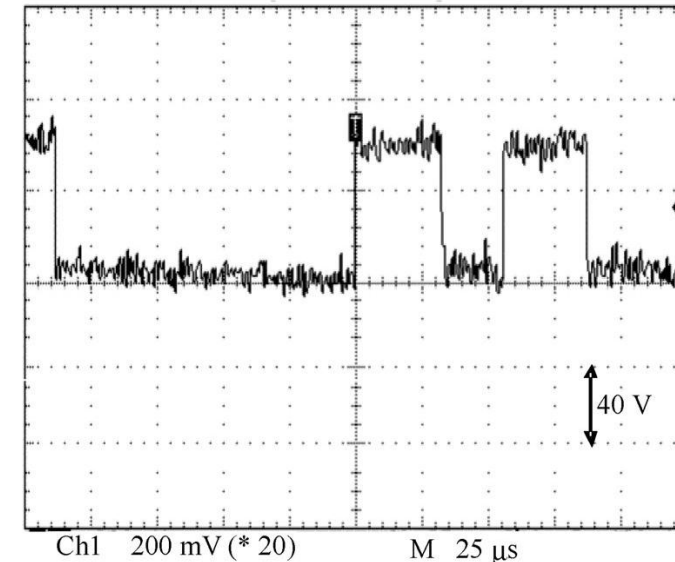
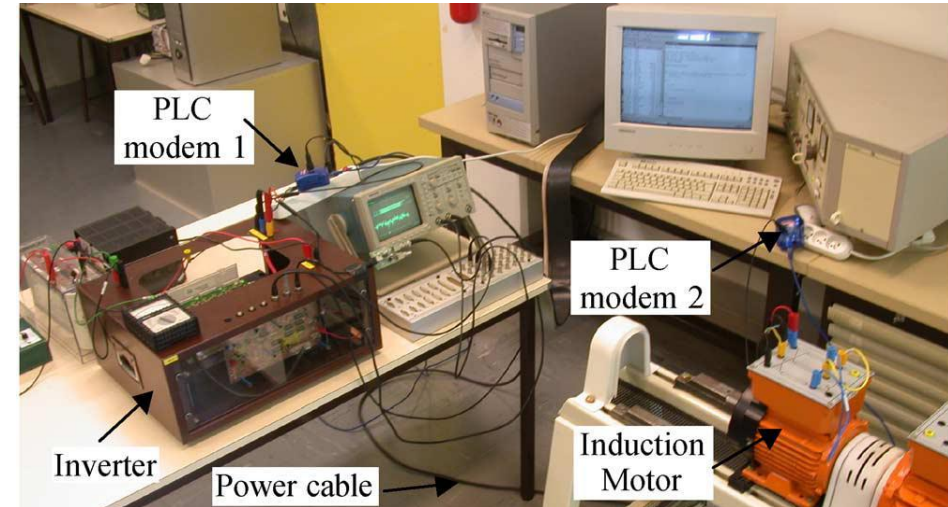
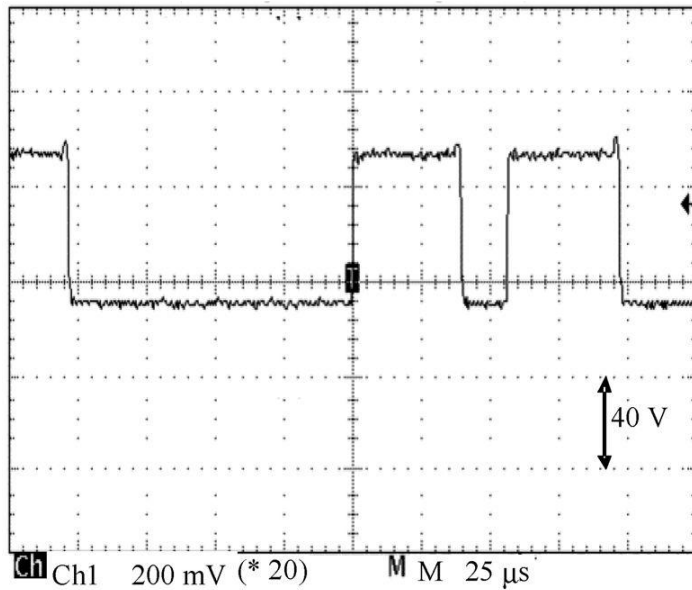
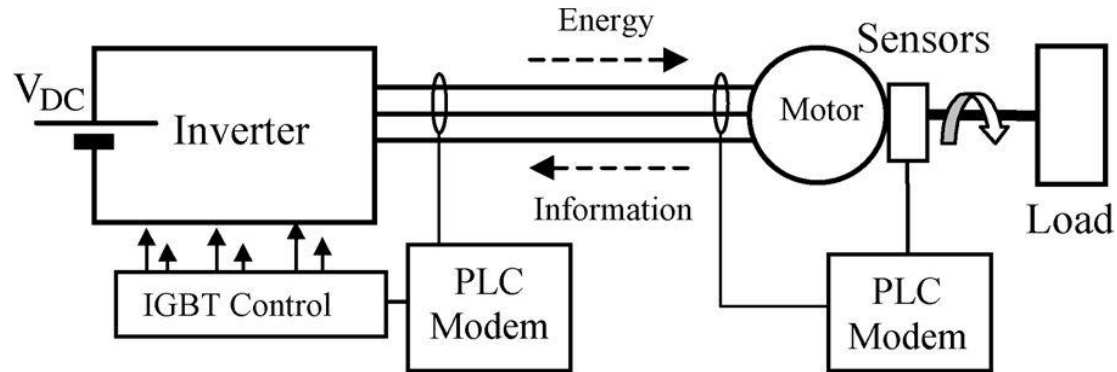


Industrial Applications

The main problem – heavy machinery, which causes interference.

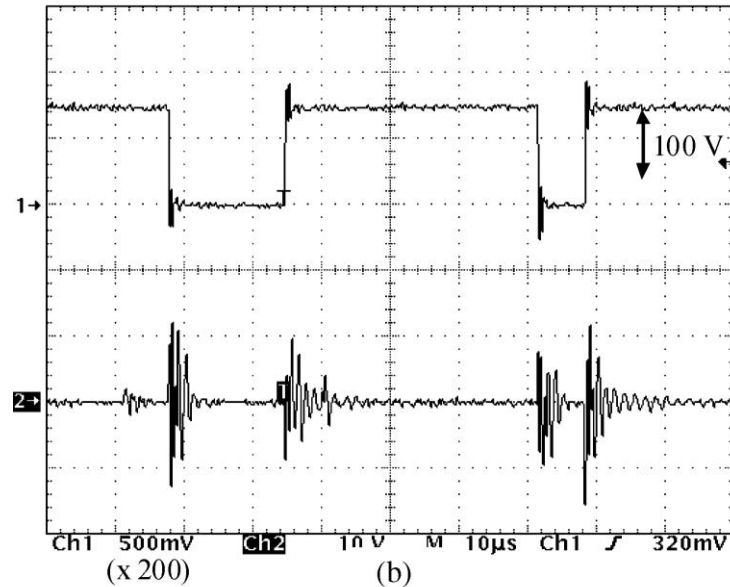
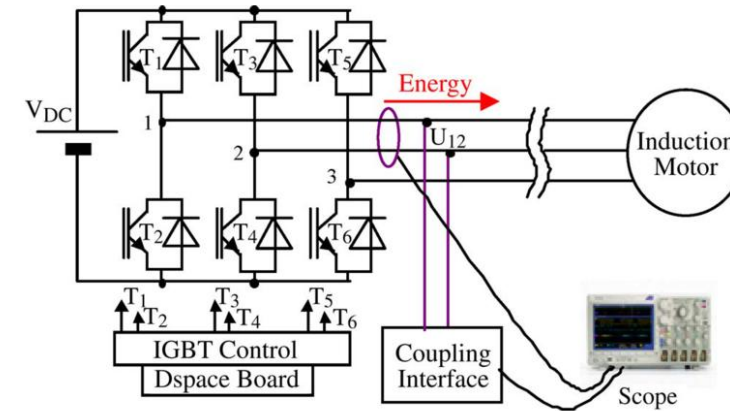
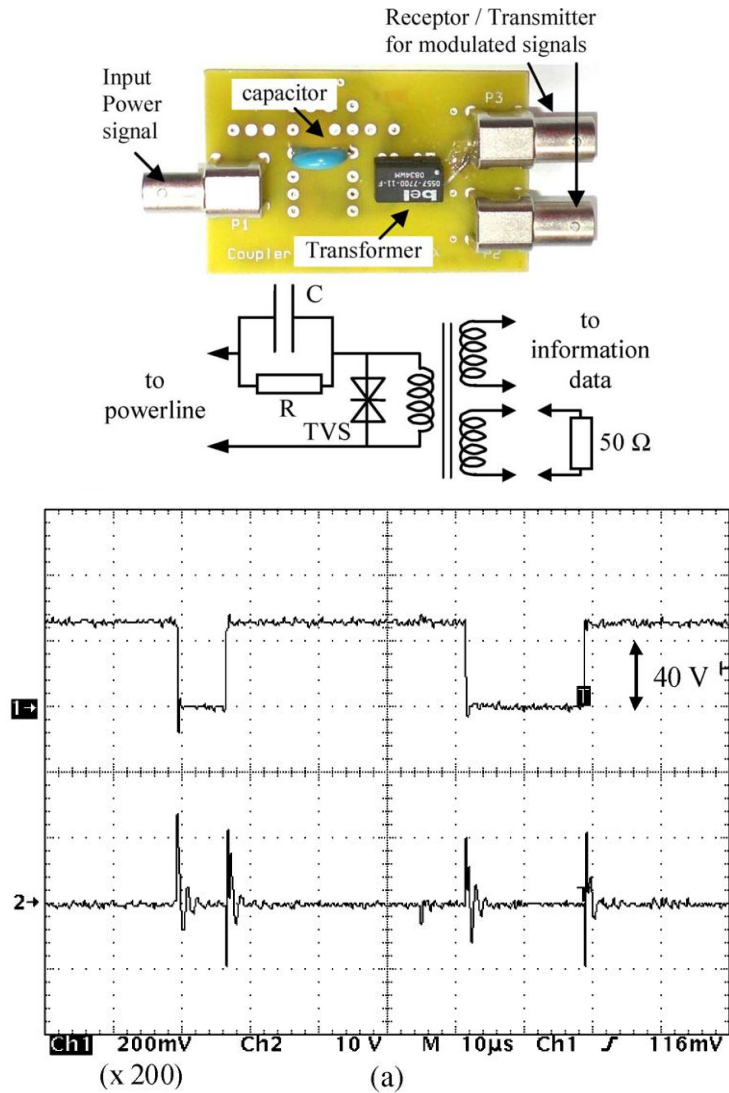


Case Study – PWM Motor Loop Back



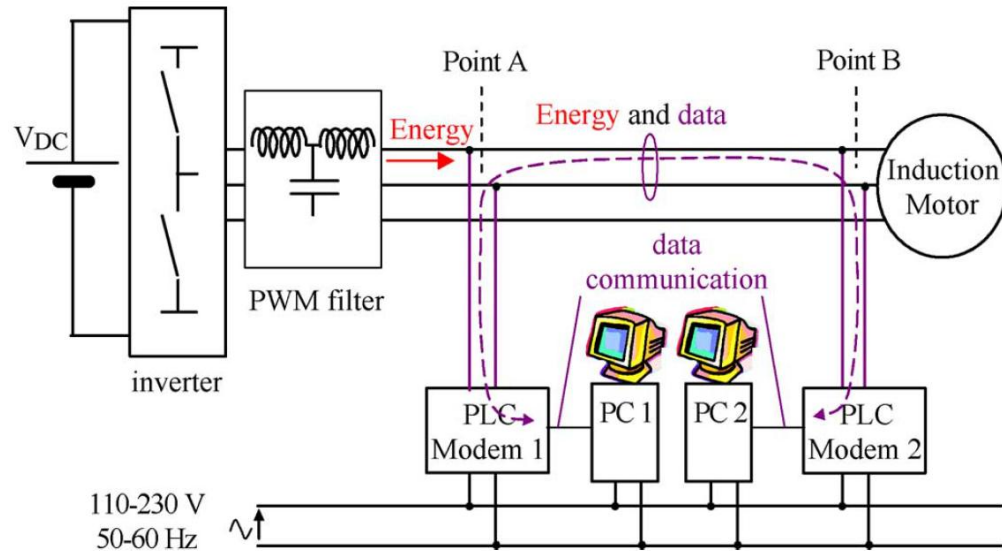
Test condition: $V_{DC}=60V$, $V_m=0.8V$, $f_m=50Hz$, $f_s=2kHz$

Case Study – PWM Motor Loop Back



Test condition: $V_m=0.8V$, $f_m=50Hz$, $f_s=2kHz$,
 (a) $V_{DC}=50V$ (b) $V_{DC}=150V$

Industrial Applications



And even though authors manage to make it work, the decrease of the data rate with higher PWM operating voltages and load current still significant.

TABLE II
DATA RATE DEPENDING ON THE VDC SET VALUE AND THE TYPE OF FILTER IN USE

V_{DC} (V)	0	50	100	150	200	250	300
Data Rate L Filter	85	33	26	24	20	19	10
Data Rate LC Filter	85	42	32	32	32	32	32
Data Rate LCL Filter	85	49	37	37	37	37	37

TABLE III
CURRENT EFFECT ON COMMUNICATION

I_M (A)	1.7	2.96	4.8	6.9	10
P_{Motor} (W)	216	531	913	1280	1740
Rate (Mbits/s)	32	29	26	23	16



Thank You!